

# IILM UNIVERSITY

## BeeSkilled Summer Internship – Machine Learning & AI

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25SCS1003003711 SEM-3

BTECH AI&ML 2CSE12



# About BeeSkilled

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- ❖ Skill development and internship platform
- ❖ Provides hands-on industry experience
- ❖ Focuses on practical, project-based learning
- ❖ Offers training in AI, ML, Data Science & other technologies
- ❖ Helps students develop technical and problem-solving skills

# Internship Overview

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- ❖ **Domain:** AI & Machine Learning
- ❖ **Duration:** 6 Weeks
- ❖ **Projects:** 3 Minor Projects and 1 Major project + PPT + Report
- ❖ **Focus:** Machine Learning & AI
- ❖ **Goal:** Gain practical ML experience

# Tools & Technologies

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- ❖ Python
- ❖ Anaconda Jupyter Notebook
- ❖ Pandas & NumPy
- ❖ Matplotlib & Seaborn
- ❖ Scikit-learn
- ❖ Machine Learning Algorithms

# Week 1-Data Preprocessing

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## Topics Learned

- ❖ Introduction to Machine Learning
- ❖ Types: Supervised, Unsupervised & Reinforcement Learning
- ❖ Importing and exploring datasets
- ❖ Handling missing data
- ❖ Encoding & Feature Scaling
- ❖ Train/Test Split

# Week 1-Data Preprocessing

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## Assignments

- ❖ Loaded datasets using Pandas
- ❖ Used .info() and .describe() for analysis
- ❖ Handled missing values using Mean/Median Imputation
- ❖ Encoded categorical data using LabelEncoder & OneHotEncoder

# Week 1-Data Preprocessing

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## Mini Project 1

### Titanic Survival Prediction – Data Cleaning Project

- ❖ Cleaned missing data
- ❖ Encoded Sex and Embarked
- ❖ Visualized age distribution
- ❖ Exported cleaned dataset as a **CSV file**

Project Link : [View link](#)

# Week 2—Model building and Evaluation

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## Topics Learned

- ❖ Linear Regression
- ❖ Logistic Regression
- ❖ Decision Tree
- ❖ Random Forest
- ❖ Evaluation Metrics: MSE,  $R^2$  & Confusion Matrix



# Week 2—Model building and Evaluation

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## Assignments

- ❖ Built Linear Regression model for housing prices
- ❖ Applied Logistic Regression for Titanic survival prediction
- ❖ Learned model training and evaluation

# Week 2—Model building and Evaluation

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## Mini Project 2

### House Price Prediction Model

- ❖ Trained Linear Regression model
- ❖ Predicted house prices
- ❖ Evaluated using  $R^2$  Score
- ❖ Plotted Predicted vs Actual Values

Project Link : [View link](#)

# Week 3-Unsupervised Learning + Model Evaluation

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## Topics Learned

- ❖ K-Means Clustering
- ❖ PCA (Principal Component Analysis)
- ❖ Feature Scaling
- ❖ Elbow Method for selecting K
- ❖ Saving/Loading models using Joblib & Pickle

# Week 3-Unsupervised Learning + Model Evaluation

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## Mini Project 3

### Iris Flower Clustering Project

- ❖ Applied K-Means (K=3)
- ❖ Visualized clusters using scatter plot
- ❖ Compared predicted clusters with true labels

**Skill Gain:** Unsupervised ML + Data Visualization

**Project Link:** [View link](#)

# Week 4-Major Capstone Project

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Major Project : Heart Disease Prediction

ML Workflow:

- ❖ Data Cleaning & Feature Scaling
- ❖ Train Logistic Regression / Decision Tree
- ❖ Evaluate Accuracy, Precision & Recall
- ❖ Confusion Matrix & ROC Curve

# Week 4-Major Capstone Project

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## Methodology

Dataset → Preprocessing → Model Training → Evaluation → Prediction

- ❖ **Preprocessing:** Cleaning, Encoding & Scaling
- ❖ **Models:** Logistic Regression & Decision Tree
- ❖ **Evaluation:** Accuracy, Precision, Recall, Confusion Matrix & ROC Curve
- ❖ **Output:** Heart Disease Prediction

# Week 4-Major Capstone Project

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## Outcome

Developed a Machine Learning model to predict the likelihood of heart disease.

## Deliverables

Working ML Model

Project Link: [View link](#)

# Skills and Learning Outcomes

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- ❖ Data Cleaning & Preprocessing
- ❖ Data Visualization
- ❖ Supervised & Unsupervised ML
- ❖ Model Training & Evaluation
- ❖ Feature Scaling & Encoding
- ❖ Practical Problem Solving
- ❖ ML Project Documentation



# Internship Outcomes

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- ❖ Completed a 6-week AI & ML internship
- ❖ Successfully completed 3 Minor Projects + 1 Major Project
- ❖ Gained hands-on experience with the ML workflow
- ❖ Improved Python and data analysis skills
- ❖ Prepared a Major Project Report & Presentation

# Project Work & Resources

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Access my internship work:

Minor Project 1: [Open](#)

Minor Project 2: [Open](#)

Minor Project 3: [Open](#)

Major Project + PPT + Report : [Open](#)

Platform: Anaconda Jupyter Notebook

# Datasets

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Access the datasets used :

[Titanic Dataset](#)

[House Prices Dataset](#)

[Iris Dataset](#)

[Heart Disease UCI Dataset](#)

# INTERNSHIP CERTIFICATE



This internship program certificate is proudly awarded to

*Tejasvi Singh*

has completed the 6-Week online internship program in **Machine Learning & AI** at

**BeeSkilled**



*S Kour*

**Simranjeet Kour**

Founder & Program Head



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**28 August 2026**

# Thank you

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